

Challenge (Habanero)

- Q1.** Factor the expression $x^3 + 8$:
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x - 2)(x^2 + 2x + 4)$
(C) $(x + 2)(x^2 + 2x + 4)$
(D) $(x - 2)(x^2 - 2x + 4)$
(E) $(x + 2)(x^2 - 2x - 4)$
- Q2.** What is the factored form of $27x^3 - 64y^3$?
(A) $(3x - 4y)(9x^2 + 12xy + 16y^2)$
(B) $(3x + 4y)(9x^2 - 12xy + 16y^2)$
(C) $(3x - 4y)(9x^2 - 12xy + 16y^2)$
(D) $(3x + 4y)(9x^2 + 12xy + 16y^2)$
(E) $(3x - 4y)(9x^2 + 12xy - 16y^2)$
- Q3.** A gamer earns $x^3 + 125$ points in a game. Factor the expression to find the number of points:
(A) $(x + 5)(x^2 - 5x + 25)$
(B) $(x - 5)(x^2 + 5x + 25)$
(C) $(x + 5)(x^2 + 5x + 25)$
(D) $(x - 5)(x^2 - 5x + 25)$
(E) $(x + 5)(x^2 - 5x - 25)$
- Q4.** Factor $8x^3 - 1$:
(A) $(2x - 1)(4x^2 + 2x + 1)$
(B) $(2x + 1)(4x^2 - 2x + 1)$
(C) $(2x - 1)(4x^2 - 2x + 1)$
(D) $(2x + 1)(4x^2 + 2x + 1)$
(E) $(2x - 1)(4x^2 + 2x - 1)$
- Q5.** What is the factored form of $x^3 - 64$?
(A) $(x - 4)(x^2 + 4x + 16)$
(B) $(x + 4)(x^2 - 4x + 16)$
(C) $(x - 4)(x^2 - 4x + 16)$
(D) $(x + 4)(x^2 + 4x + 16)$
(E) $(x - 4)(x^2 + 4x - 16)$
- Q6.** Factor $27x^3 + 8y^3$:
(A) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
(B) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
(C) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
(D) $(3x - 2y)(9x^2 - 6xy + 4y^2)$
(E) $(3x + 2y)(9x^2 - 6xy - 4y^2)$
- Q7.** A coding error occurs when $x^3 - 1000$ is not factored. Factor the expression:
(A) $(x - 10)(x^2 + 10x + 100)$
(B) $(x + 10)(x^2 - 10x + 100)$
(C) $(x - 10)(x^2 - 10x + 100)$
(D) $(x + 10)(x^2 + 10x + 100)$
(E) $(x - 10)(x^2 + 10x - 100)$
- Q8.** The high score in a game is given by $x^3 + 27$. Factor the expression:
(A) $(x + 3)(x^2 - 3x + 9)$
(B) $(x - 3)(x^2 + 3x + 9)$
(C) $(x + 3)(x^2 + 3x + 9)$
(D) $(x - 3)(x^2 - 3x + 9)$
(E) $(x + 3)(x^2 - 3x - 9)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $x^3 - 216$ and explain how it can be used in a gaming application.
- Q10.** A gaming company wants to factor $8x^3 + 1$ to optimize data usage. Factor the expression and provide a brief explanation of your method.

Chef's Tips

- Chef's Tips: Use the sum and difference of cubes formulas to factor expressions.
- Chef's Tips: Consider the context of tech and gaming when creating word problems.
- Chef's Tips: Encourage students to use factoring to solve complex problems.